

Loans and Credit

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1. Introduction

The key drivers of default risk are explored. The following factors of individuals are considered: whether the income source was verified, the debt-to-income ratio, the number of year that the individual has had any sort of credit line, prior bankruptcies, the number of mortgage accounts, bankcard utilisation, interest rate, term length, and history of credit behaviour.

1.1. The Key Risk Drivers for Default

A logistic regression is used to study the determinants of default risk, as the dependent variable is binary in nature. Across both specifications, all the coefficients are statistically significant, justifying their statuses as key drivers of default risk. The Lending Club's credit grade system proves to be great indicator of debt default, with lower grades being associated with an increased probability of default. The DTI ratio, prior bankruptcies, and bankcard utilisation are also associated with increased default risk. The number of mortgage accounts and never having missed payments on debt are associated with lower default risk.

```
##
## Logistic Regression: Predictors of Loan Default
## =====
##                               Dependent variable:
##                               -----
##                               Default (1 = Loss, 0 = Paid)
##                               (1)           (2)
## -----
## Source Verified              0.346***      0.188***
##                               (0.010)      (0.011)
```

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## Verified	0.555***	0.283***
##	(0.011)	(0.012)
## DTI	0.027***	0.015***
##	(0.0005)	(0.0005)
## Credit Age	-0.006***	0.004***
##	(0.001)	(0.001)
## Prior Bankruptcies	0.179***	0.079***
##	(0.009)	(0.010)
## % Never Delinquent	-0.007***	-0.004***
##	(0.0004)	(0.0005)
## Mortgage Accounts	-0.126***	-0.123***
##	(0.003)	(0.003)
## Bankcard Utilisation	0.005***	0.002***
##	(0.0001)	(0.0002)
## No. Revolving Accounts	0.005***	0.005***
##	(0.001)	(0.001)
## Grade B		0.655***
##		(0.020)
## Grade C		1.044***
##		(0.026)
## Grade D		1.290***
##		(0.037)
## Grade E		1.413***
##		(0.048)
## Grade F		1.590***
##		(0.060)
## Grade G		1.678***
##		(0.078)
## Interest Rate		0.028***
##		(0.003)
## Term		0.016***
##		(0.0004)
## Constant	-1.409***	-3.419***
##	(0.047)	(0.056)
## -----		
## Observations	372,684	372,684
## Log Likelihood	-191,565.700	-180,892.700
## Akaike Inf. Crit.	383,151.300	361,821.500
## =====		

Note: *p<0.05; **p<0.01; ***p<0.001

1.2. Default-Risk Customer Profiles

1.2.1. Default Rate by Lending Club's Customer Grading System

The plot below illustrates that default rates increase as credit grades worsen. Customers that are assigned poor Lending Club credit grades are at higher risk of defaulting on their debt.

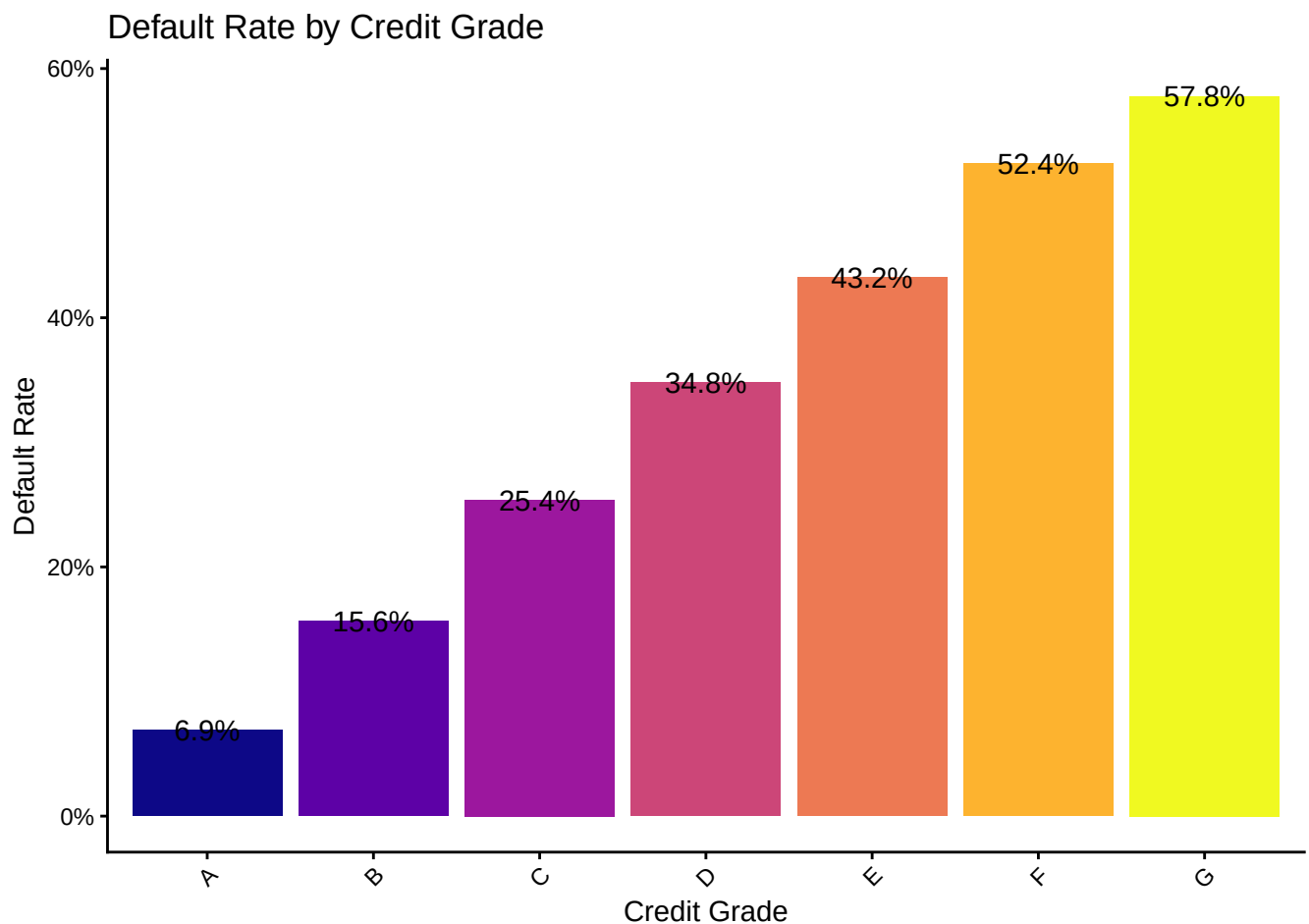


Figure 1.1: Default Rate by Grading System .

1.2.2. Default Rate by Home Ownership

Individuals that rent instead of own their house/mortgage have an average default rate of 26.9%. Owning a mortgage is associated with the lowest average default rate, supporting the findings of the logit regression, where the number of mortgages negatively relates to the default rate. This may

simply be a proxy for income.

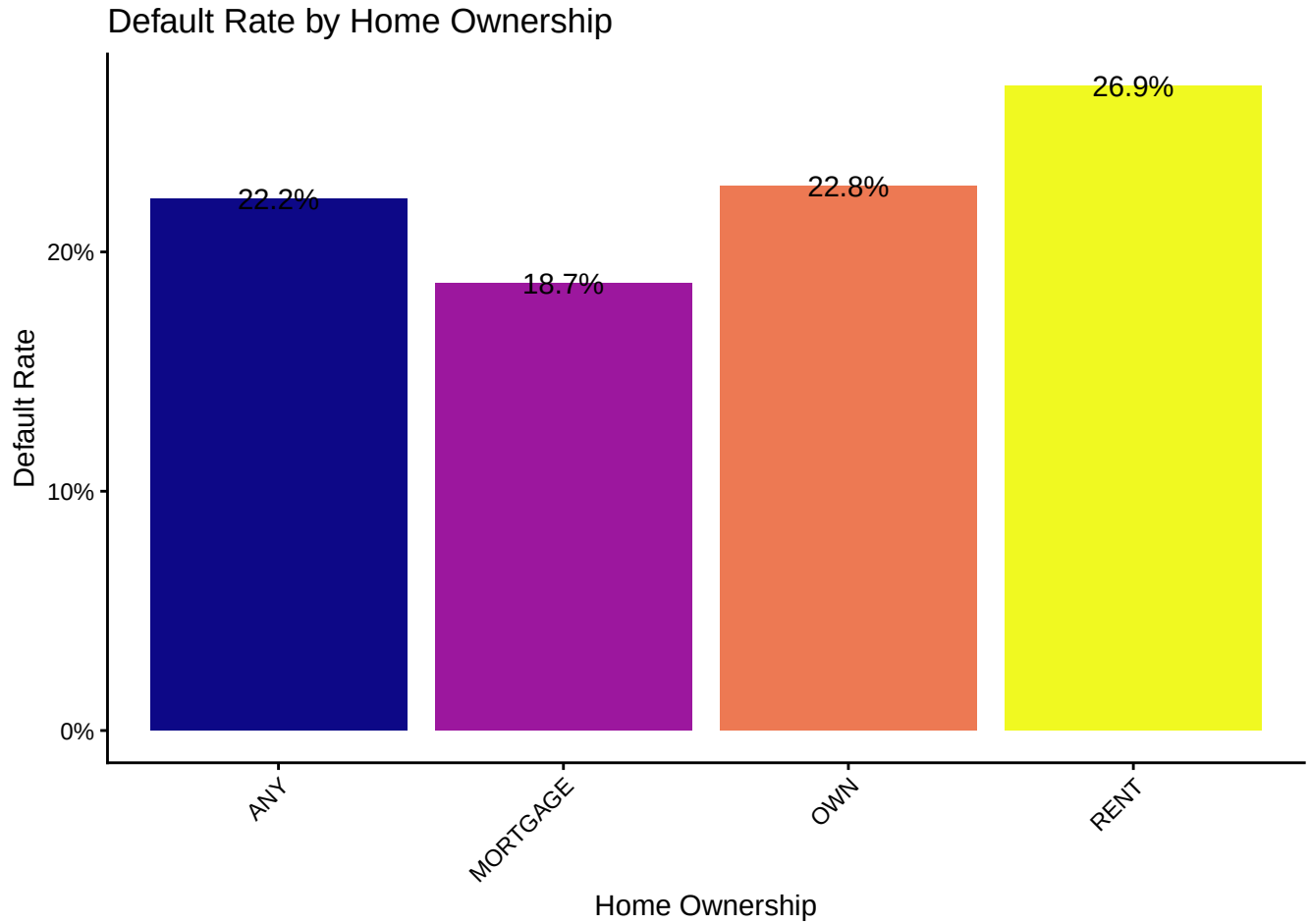


Figure 1.2: Default Rate by Home Ownership.

1.2.3. Default Rate by Employment Length

The default rate appears to be similarly distributed across employment lengths, with a slight downward trend, such that longer employment years (stable income) indicate a slightly lower risk of default.

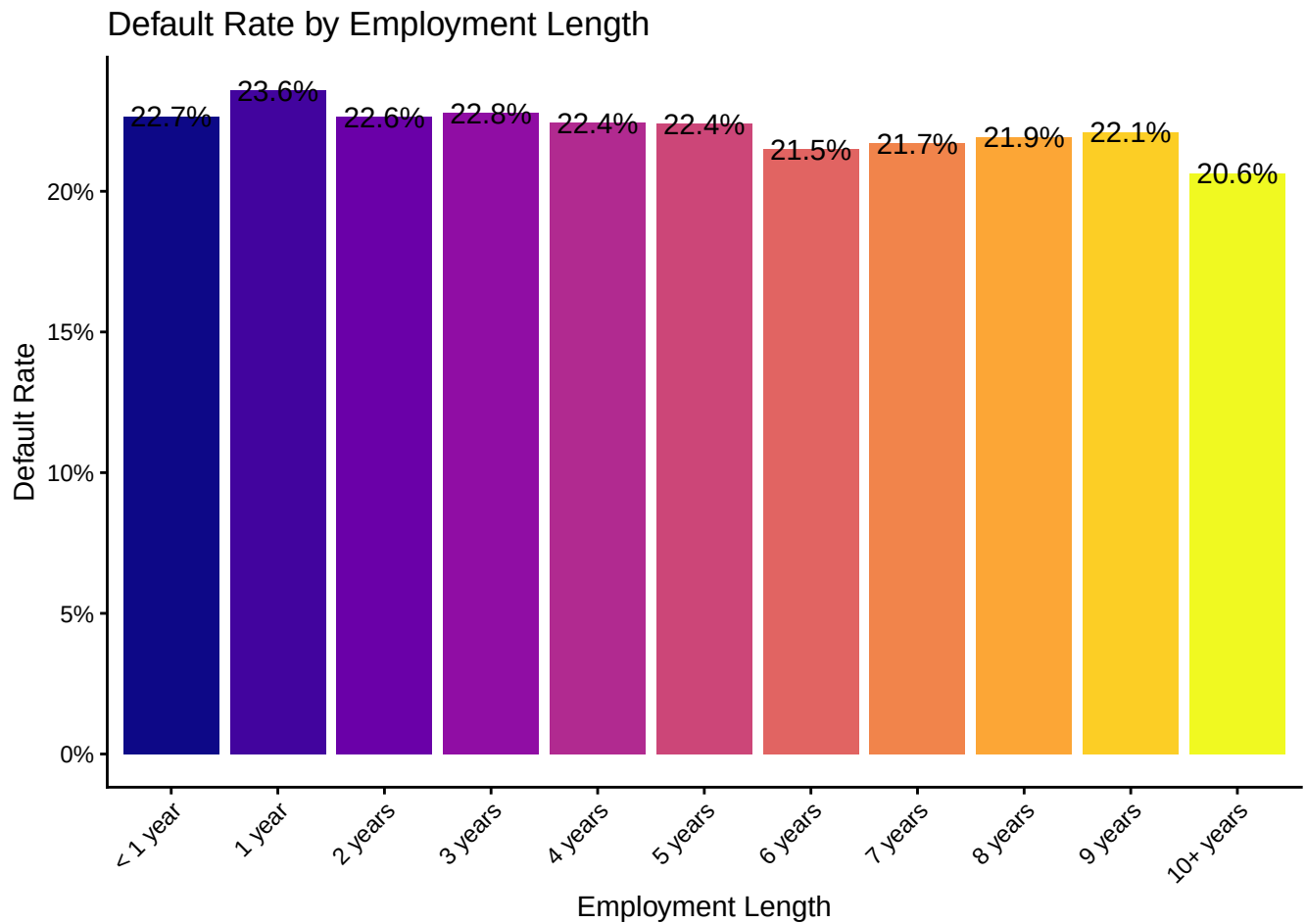


Figure 1.3: Default Rate by Employment Length.

1.2.4. Default Rate by State for Both Long and Short Term Loans

Arkansas, Los Angeles, Mississippi, Nebraska, and Oklahoma have the highest average default rates, signalling that individuals from these states introduce increased risk. Whereas individuals from DC, Maine, New Hampshire, Oregon, Vermont, and West Virginia signal prudence, as these states have the lowest average default rates.

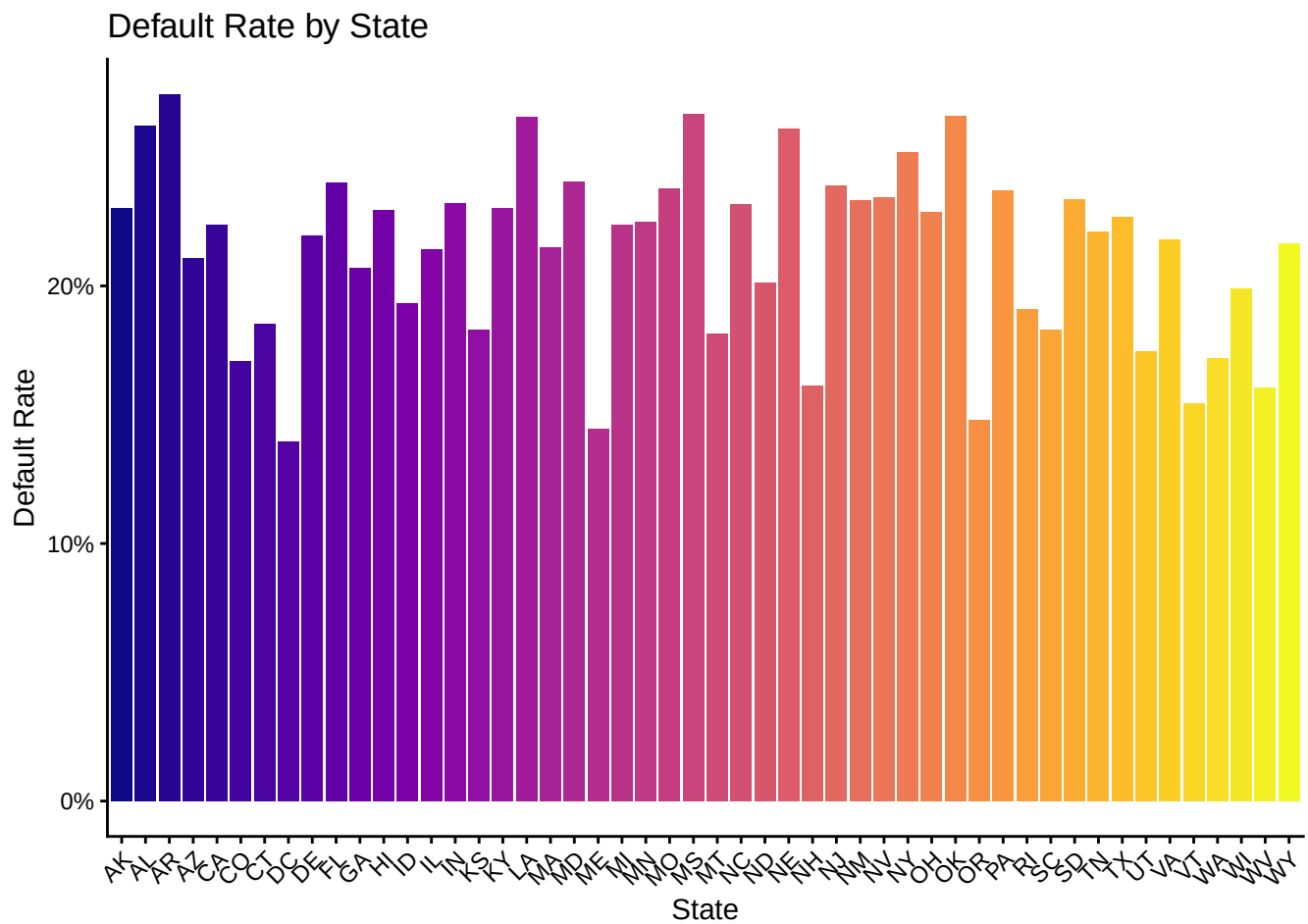


Figure 1.4: Default Rate by State for Both Long and Short Term Loans.

1.2.5. Default Rate by State for Short Term Loans Only

The state analysis is similar when only short-term loans are considered.

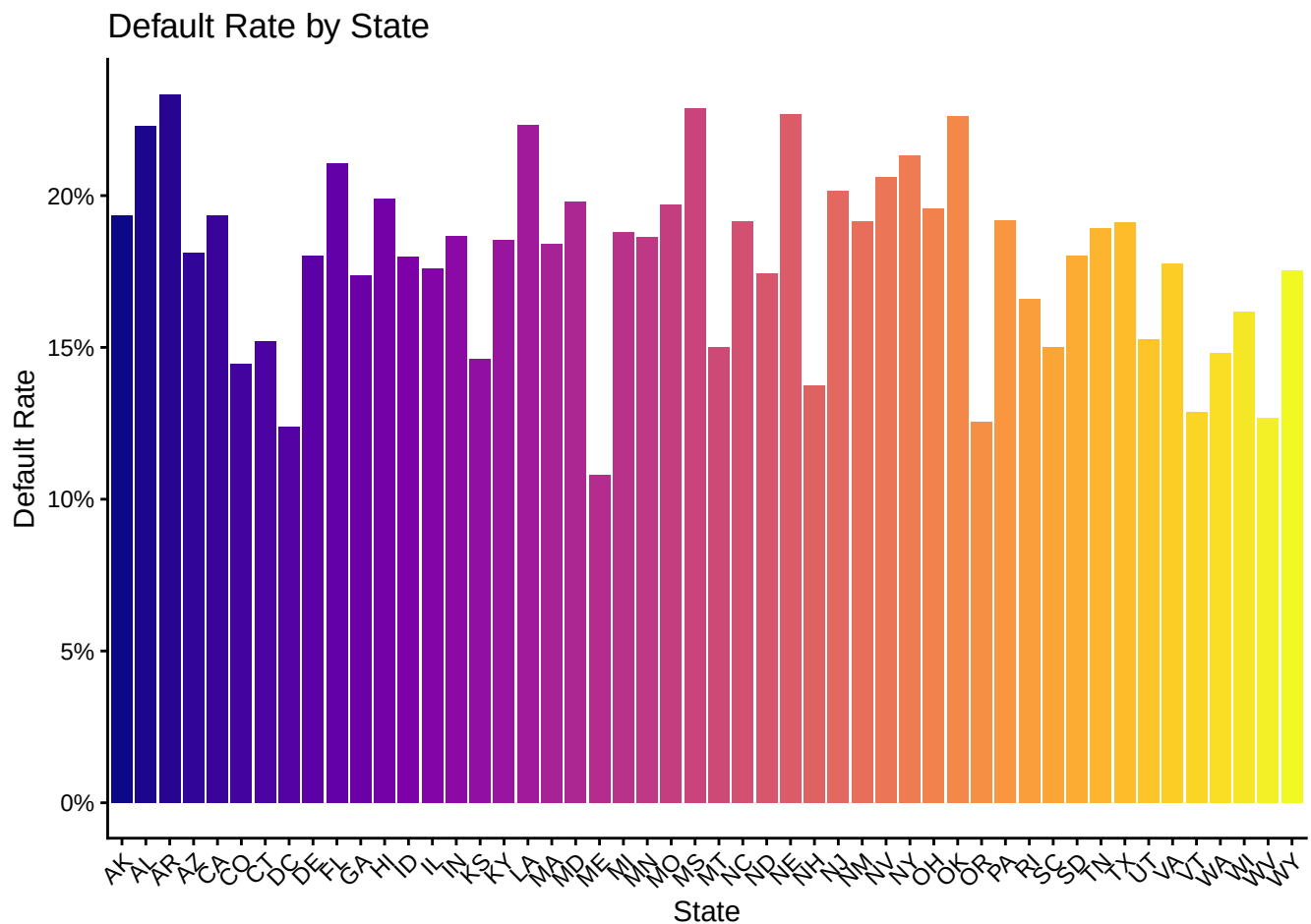


Figure 1.5: Default Rate by State for Short Term Loans Only.

1.2.6. Default Rate by Purpose of Loan

Individuals who cite funding a small business or planning a wedding as the reason for their debt obligations are at the highest risk of defaulting on their loans.

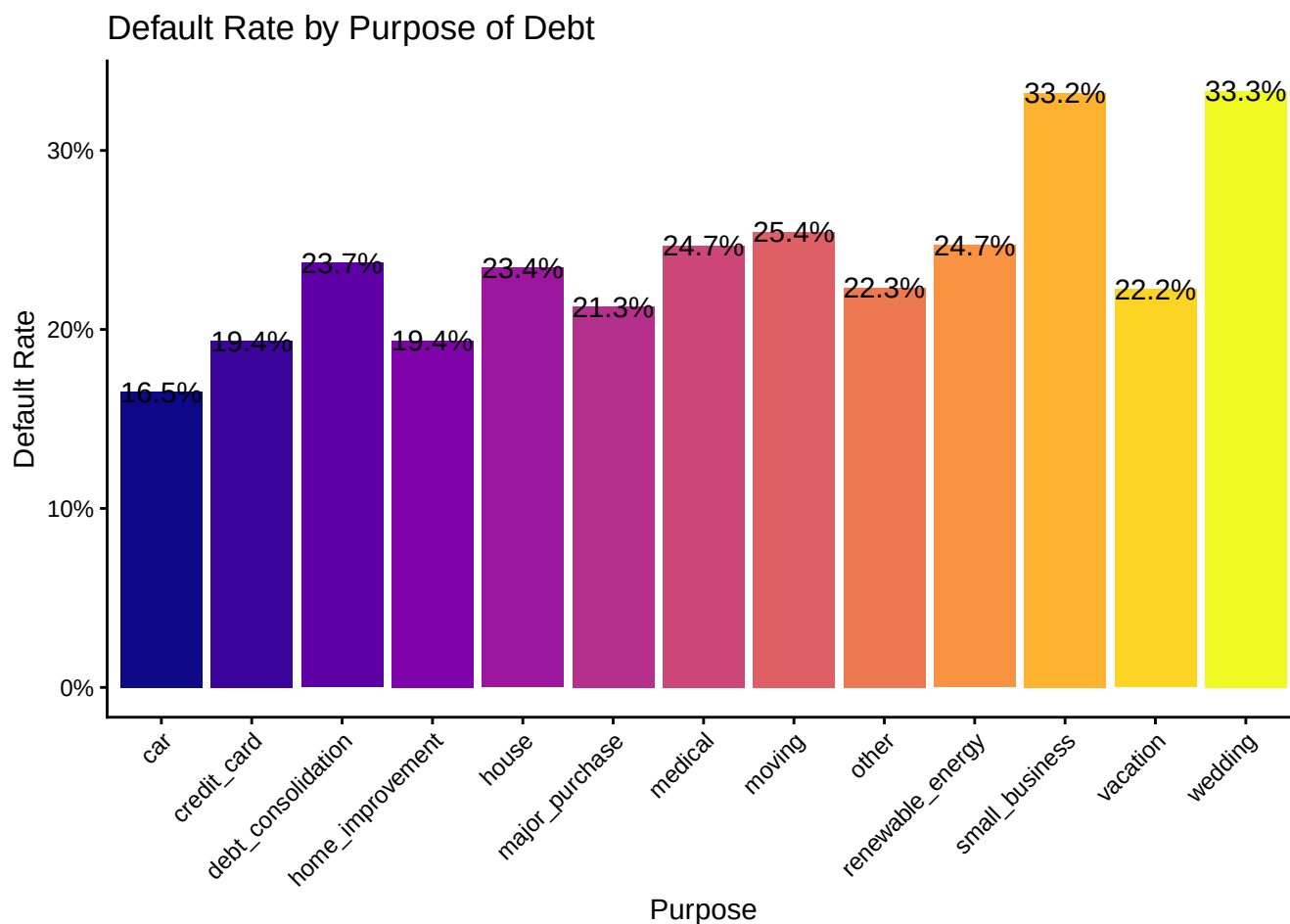


Figure 1.6: Default Rate by Purpose.

1.3. The Debt-to-Income and Loan Status Relationship

The average debt-to-income ratio is the highest for those who have defaulted on their loans and completely written their loans off, although the variance is not large across these broader loan status groups.

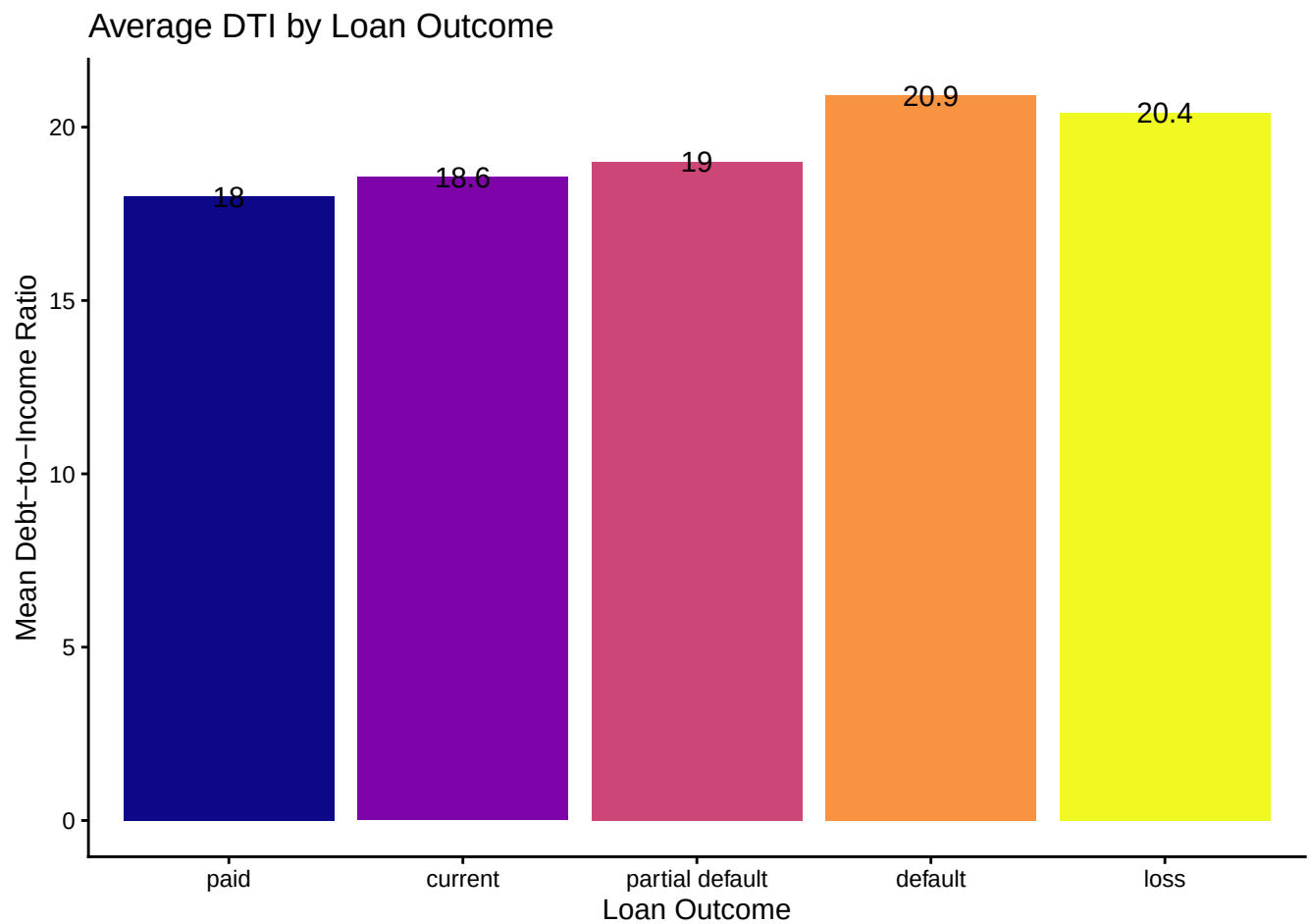


Figure 1.7: Average DTI by Loan Status.

2. Conclusion

Multiple factors must be considered when approving loan applications to ensure that the default rate is minimised. By scrutinizing over where individuals reside, the debt-to-income ratios, the purpose of the credit, the credit category, home ownership, length of employment, debt history, and bankcard utilisation, debt default risk can be improved.